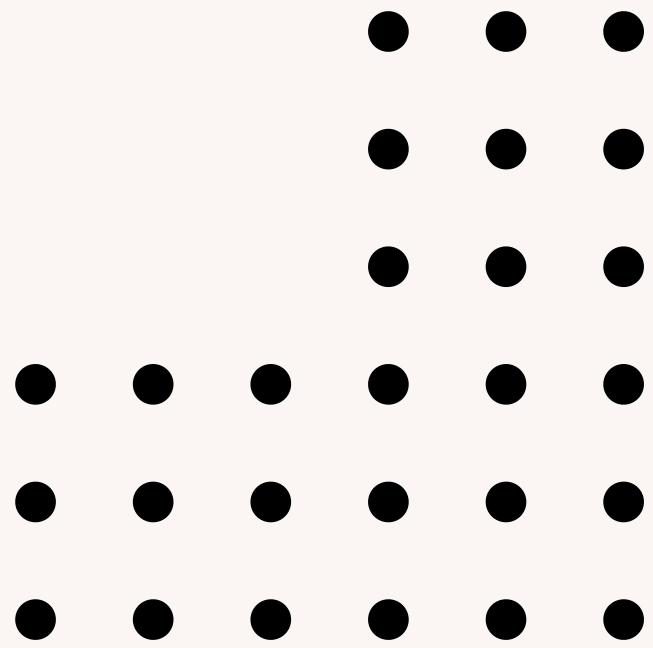




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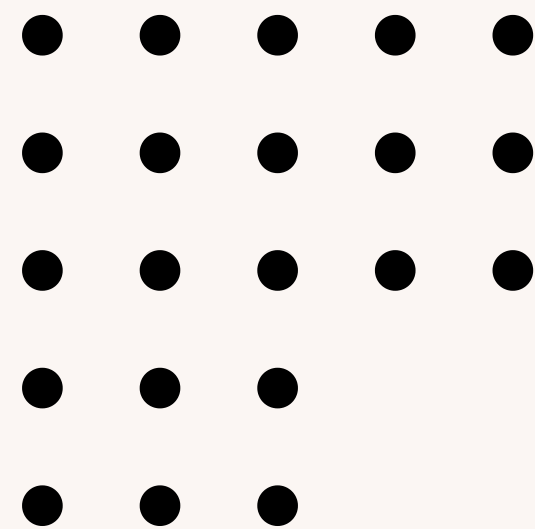
Material Library in GLSL (Nau)

Grupo 08:

Ricardo Sousa (pg60299)

Pedro Gonçalves (pg60290)

Projeto em Computação Gráfica e Visão por Computador 2025/2026



TOPICS

① The Original Cook–Torrance Model

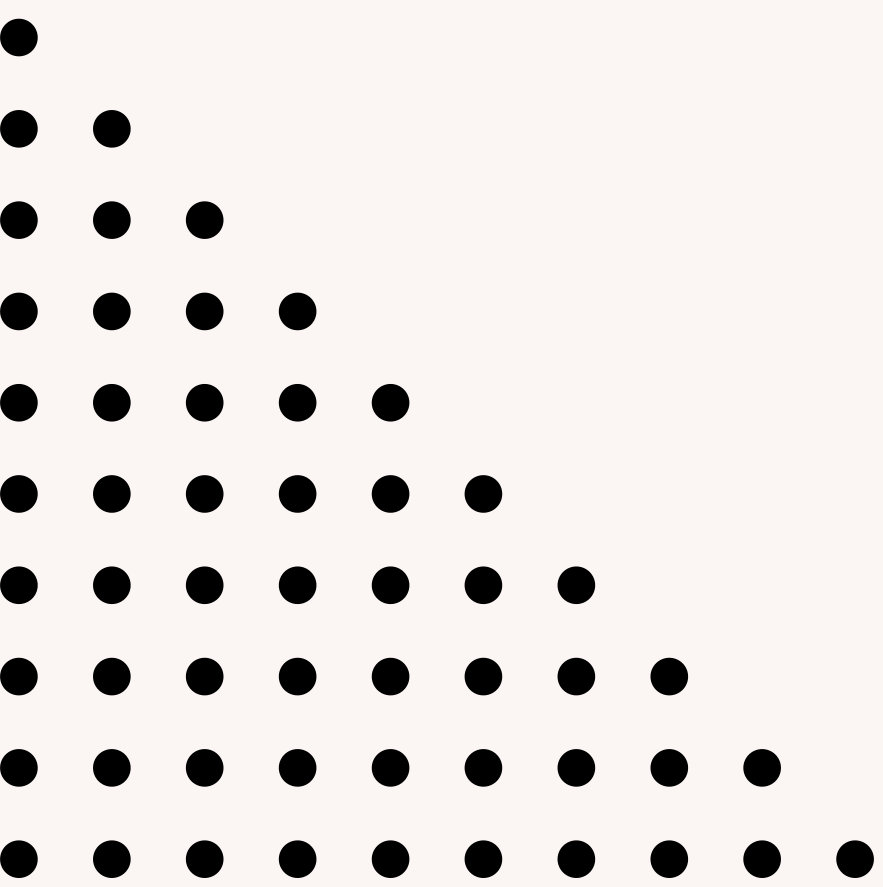
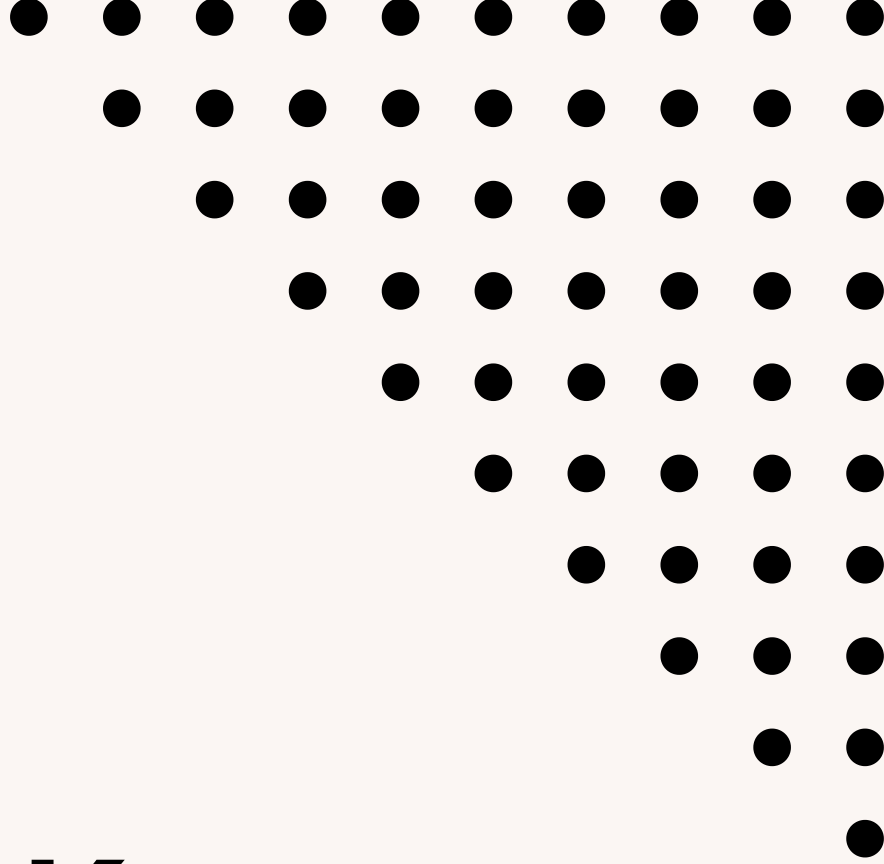
- ①.1 Formula
- ①.2 Fresnel Function
- ①.3 Beckmann Normal Distribution Function
- ①.4 Geometry Function

② The Modern Cook–Torrance Model

- ②.1 Formula
- ②.2 fresnel-schlick Function
- ②.3 Trowbridge-Reitz GGX Normal Distribution Function
- ②.4 Geometry Function

TOPICS

- ③ Visual Differences
 - ③.1 Distribution Function
 - ③.2 Geometry Function



1. THE ORIGINAL COOK TORRANCE MODEL

1.1 FORMULA

$$R_s = \frac{F}{\pi} \frac{D}{(N \cdot L)} \frac{G}{(N \cdot V)}$$

```
float D_val = D_Original(alpha, N, H);  
float G_val = G_Original(N, V, L, H);  
float F_val = F_Original(n_derived, V, H);  
  
float Rs_num = D_val * G_val * F_val;  
float Rs_den = PI * NdotV * NdotL;  
  
Rs = vec3(Rs_num / max(Rs_den, 0.000001));
```


1.2 FRESNEL FUNCTION

$$F = \frac{1}{2} \frac{(g-c)^2}{(g+c)^2} \left\{ 1 + \frac{(c(g+c)-1)^2}{(c(g-c)+1)^2} \right\}$$

where

$$c = \cos(\theta) = \mathbf{V} \cdot \mathbf{H}$$
$$g^2 = n^2 + c^2 - 1.$$

```
// Fresnel Function
float F_Original(float n, vec3 V, vec3 H) {
    float c = max(dot(V, H), 0.0);
    float g2 = n * n + c * c - 1.0;
    float g = sqrt(max(g2, 0.0));

    float gMinusC = g - c;
    float gPlusC = max(g + c, 0.000001);

    float termo1 = (gMinusC * gMinusC) / (gPlusC * gPlusC);

    float termo2_num = (c * gPlusC - 1.0);
    float termo2_den = (c * gMinusC + 1.0);
    float termo2 = 1.0 + (termo2_num * termo2_num) / (termo2_den * termo2_den);

    return 0.5 * termo1 * termo2;
}
```

1.3 BECKMANN NORMAL DISTRIBUTION FUNCTION

$$D = \frac{1}{m^2 \cos^4 \alpha} e^{-\left\{ \frac{\tan^2 \alpha}{m^2} \right\}}$$

```
// Beckmann Normal Distribution Function
float D_Original(float m, vec3 N, vec3 H) {
    float NdotH = clamp(dot(N, H), 0.000001, 1.0);
    float NdotH2 = NdotH * NdotH;
    float m2 = m * m;

    float tan2Alpha = max(1.0 - NdotH2, 0.0) / max(NdotH2, 0.000001);

    float numerator = exp(-tan2Alpha / m2);
    float denominator = m2 * NdotH2 * NdotH2;

    return numerator / denominator;
}
```

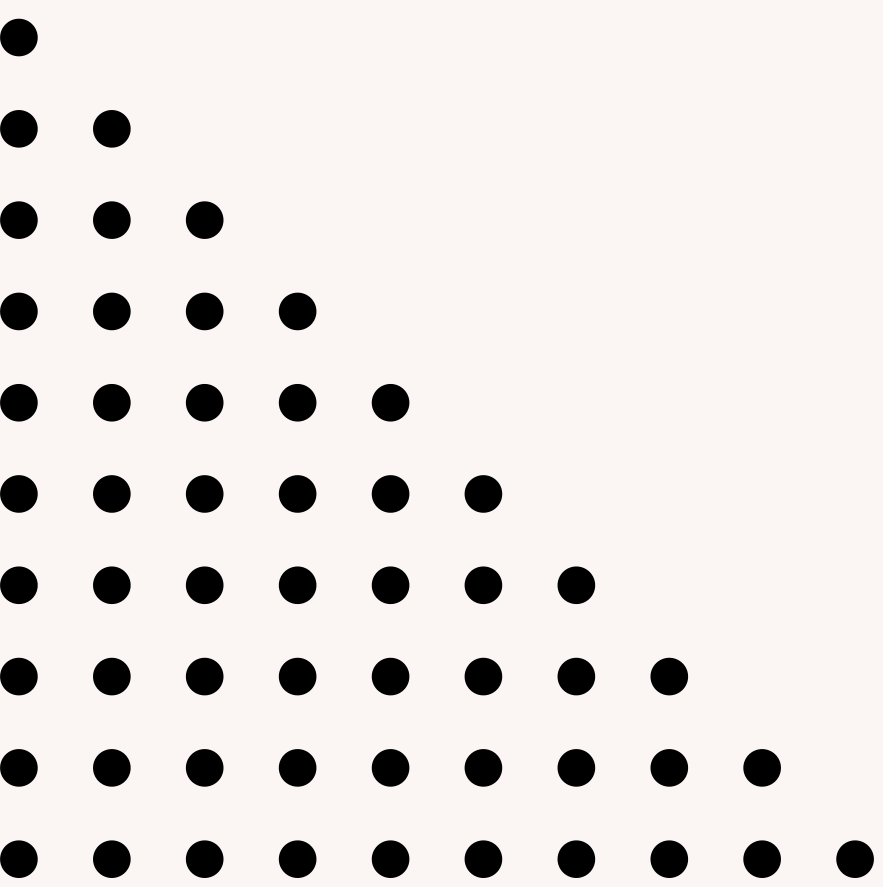
1.4 GEOMETRY FUNCTION

$$G = \min \left\{ 1, \frac{2(N \cdot H)(N \cdot V)}{(V \cdot H)}, \frac{2(N \cdot H)(N \cdot L)}{(V \cdot H)} \right\}$$

```
// Geometry Function
float G_Original(vec3 N, vec3 V, vec3 L, vec3 H) {
    float NdotH = max(dot(N, H), 0.0);
    float NdotV = max(dot(N, V), 0.0);
    float NdotL = max(dot(N, L), 0.0);
    float VdotH = max(dot(V, H), 0.000001);

    float masking = (2.0 * NdotH * NdotV) / VdotH;
    float shadowing = (2.0 * NdotH * NdotL) / VdotH;

    return min(1.0, min(masking, shadowing));
}
```

2. THE MODERN COOK

TORRANCE MODEL

2.1 FORMULA

$$f_r(\mathbf{i}, \mathbf{o}, \mathbf{n}) = \frac{F(\mathbf{i}, \mathbf{h}_r) G(\mathbf{i}, \mathbf{o}, \mathbf{h}_r) D(\mathbf{h}_r)}{4 |\mathbf{i} \cdot \mathbf{n}| |\mathbf{o} \cdot \mathbf{n}|}$$

```
float D_val = D_Modern(alpha, N, H);  
float G_val = G_Modern(N, V, L, alpha);  
vec3 F_val = F_Modern(max(dot(H, V), 0.0), F0);  
  
vec3 Rs_num = D_val * G_val * F_val;  
float Rs_den = 4.0 * NdotV * NdotL;
```

2.2 FRESNEL-SCHLICK FUNCTION

$$F_{\text{Schlick}} = F_0 + (1 - F_0)(1 - \cos \theta_d)^5$$

```
// Fresnel-Schlick
vec3 F_Modern(float cosTheta, vec3 F0) {
    return F0 + (1.0 - F0) * pow(clamp(1.0 - cosTheta, 0.0, 1.0), 5.0);
}
```

2.3 TROWBRIDGE-REITZ GGX NORMAL DISTRIBUTION FUNCTION

$$D(\mathbf{m}) = \frac{\alpha_g^2 \chi^+(\mathbf{m} \cdot \mathbf{n})}{\pi \cos^4 \theta_m (\alpha_g^2 + \tan^2 \theta_m)^2}$$

$$D(H) = \frac{\alpha^2}{\pi (\cos^2 \theta_m (\alpha^2 - 1) + 1)^2}$$

```
// Trowbridge-Reitz GGX Normal Distribution Function
float D_Modern(float roughness, vec3 N, vec3 H) {
    float a = roughness * roughness;
    float a2 = a * a;
    float NdotH = max(dot(N, H), 0.0);
    float NdotH2 = NdotH * NdotH;

    float num = a2;
    float denom = (NdotH2 * (a2 - 1.0) + 1.0);
    denom = PI * denom * denom;

    return num / max(denom, 0.000001);
}
```

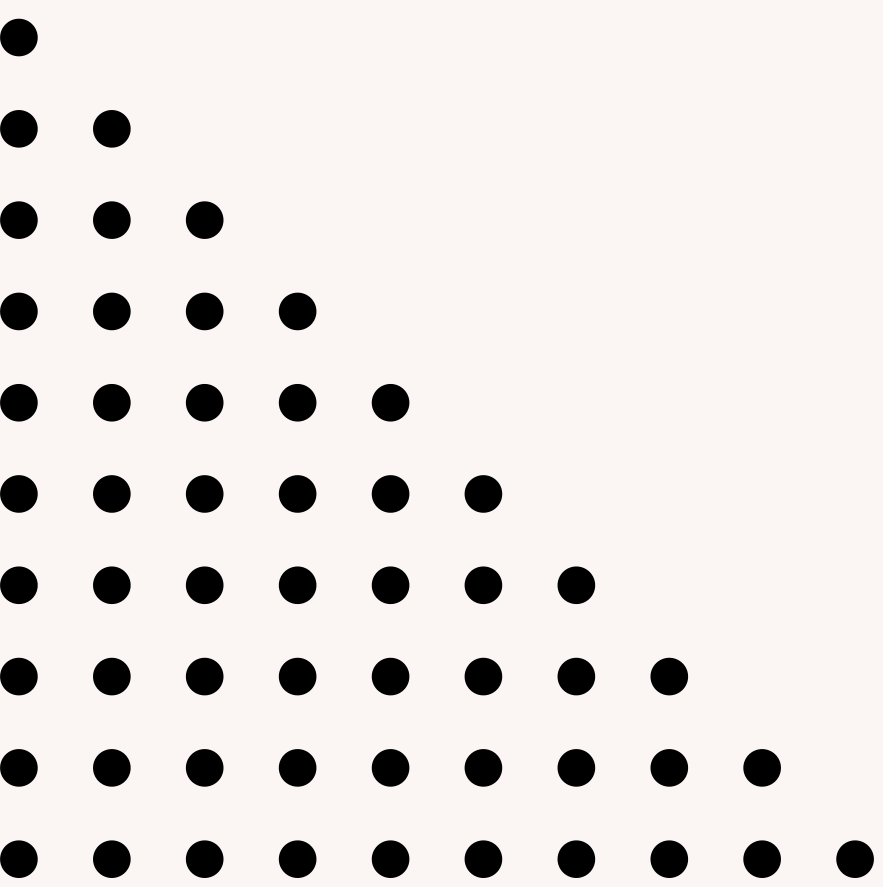
2.4 GEOMETRY FUNCTION

$$G_{GGX} = G_1(\vec{v}) * G_1(\vec{l})$$
$$G_1(x) = \chi^+ \left(\frac{\vec{x} \cdot \vec{m}}{\vec{x} \cdot \vec{n}} \right) \frac{2}{1 + \sqrt{1 + \alpha^2 \tan^2(\theta_x)}}$$

```
// Schlick-GGX Geometry Function
float GeometrySchlickGGX(float NdotV, float roughness) {
    // Para luzes diretas, mapeia-se o K assim:
    float r = (roughness + 1.0);
    float k = (r * r) / 8.0;

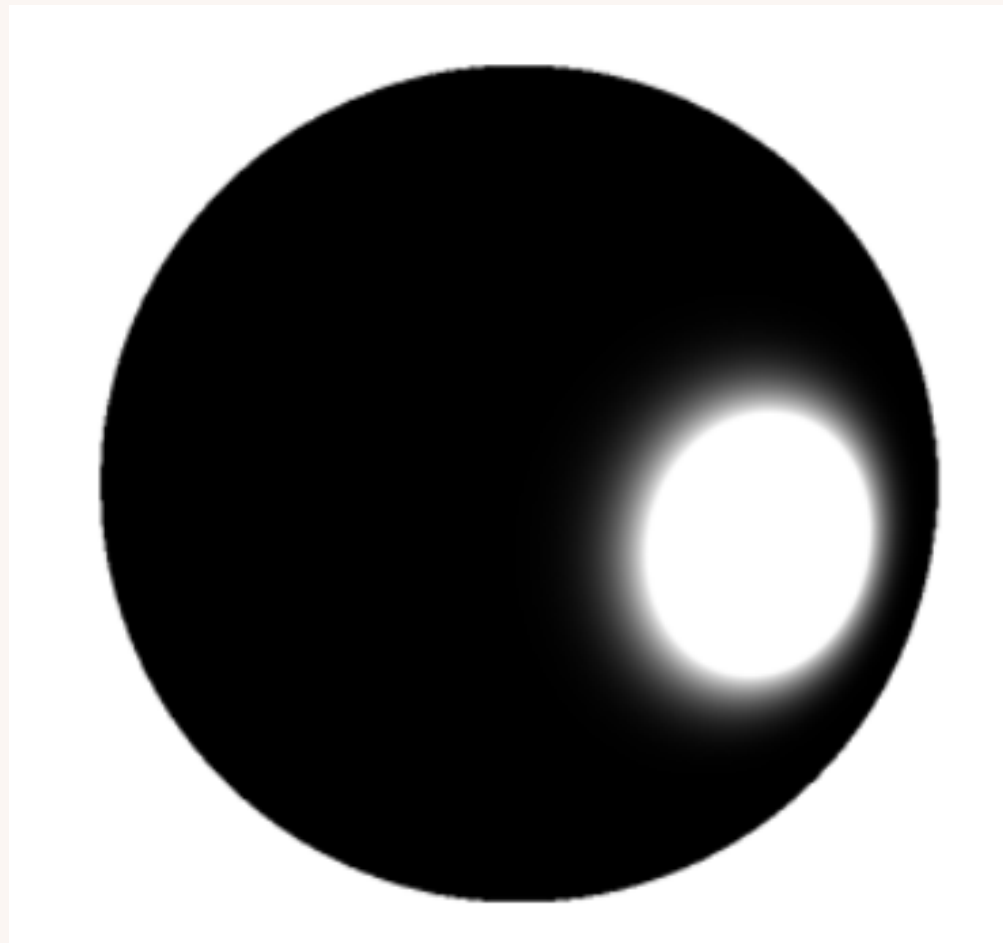
    float num = NdotV;
    float denom = NdotV * (1.0 - k) + k;
    return num / denom;
}

// Smith Geometry Function
float G_Modern(vec3 N, vec3 V, vec3 L, float roughness) {
    float NdotV = max(dot(N, V), 0.0);
    float NdotL = max(dot(N, L), 0.0);
    float ggx1 = GeometrySchlickGGX(NdotV, roughness);
    float ggx2 = GeometrySchlickGGX(NdotL, roughness);
    return ggx1 * ggx2;
}
```

3. VISUAL DIFFERENCES

3.1 DISTRIBUTION FUNCTION

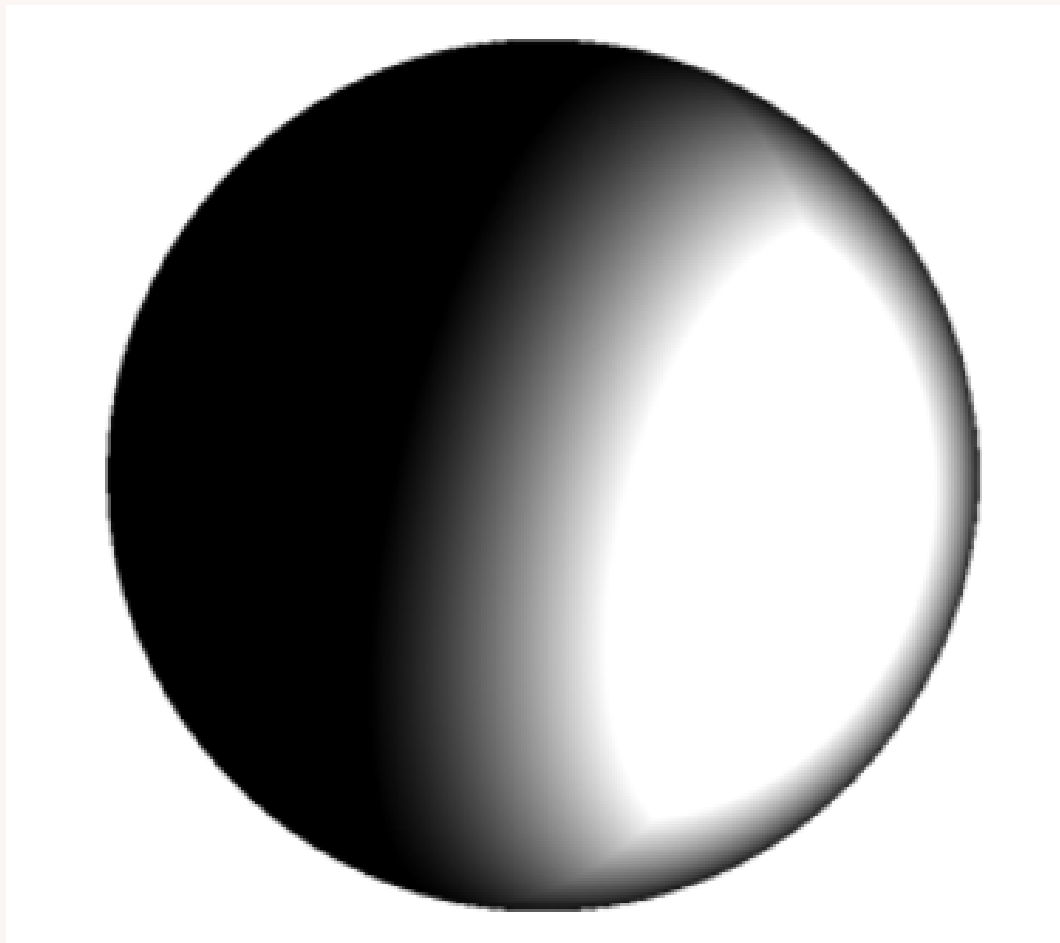


BECKMANN



GGX

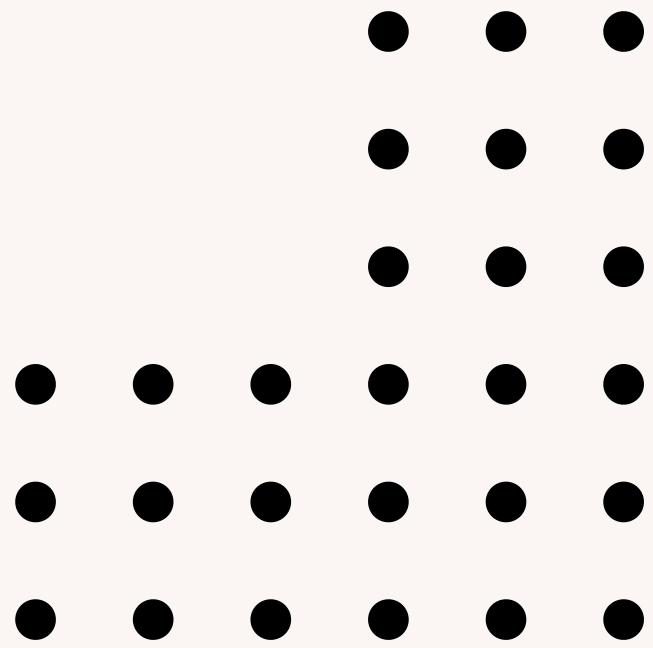
3.2 GEOMETRY FUNCTION



COOK-TORRANCE



GGX



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